



**Ahmedabad**  
**University**

CSE623 : Machine Learning

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**Machine Learning-based Estimation of Muscle  
Cross-Sectional Area (ACSA) from Segmented  
Ultrasound Images of Lower Limb Muscles for  
Sports Performance Analysis**

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## **Abstract**

We assessed the shortcomings of our ultrasound image segmentation pipeline that included boundary error and leakage of segments. Through this analysis we found that there was a necessity to have better feature representation with the use of texture, gradient as well as multiscale features and also the use of boundary sensitive methods to increase accuracy of segmentation. We have also investigated data augmentation methods and intended to optimize the Random Forest model by tuning hyperparameters and better addressing the problem of class imbalance. The future work will apply more sophisticated feature extraction algorithms, including GLCM and LBP, better post-processing strategies, and other measures of evaluation to obtain a more precise segmentation and more dependable ACSA estimation.

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# 1 What We Have Done

- We carried out a detailed evaluation of the limitations in our existing segmentation pipeline, and focused particularly on boundary inaccuracies and segmentation leakage in challenging regions.
- Working upon these observations, we identified the key areas that required improvement in feature representation, including the need for more robust texture, gradient, and multiscale features to better capture ultrasound image characteristics.
- We further performed a conceptual study of boundary refinement techniques, along with analyzing how edge-aware features and improved sampling strategies could enhance segmentation precision.
- We explored and shortlisted strategies for advanced augmentation, such as elastic deformation and noise simulation, in order to improve the robustness of the model in low-contrast and noisy ultrasound conditions.
- We also got an outline of a plan for optimizing the Random Forest Model, including identifying certain important hyperparameters(e.g., number of trees, depth, feature subsets) and better handling of class imbalance.
- In addition to that, we also reviewed our current post-processing pipeline and proposed various improvements, such as adaptive morphological operations and contour refinement, in order for us to achieve more consistent and smoother segmentation masks.
- Lastly, we examined the relationship between segmentation accuracy and ACSA estimation to better identify the need for improved calibration and reduced segmentation noise for more reliable area prediction.

## 2 What We Aspire to Do

- Next week, we are going to start working on the improved feature extraction pipeline, including the new texture descriptors, GLCM and LBP, as well as gradient based and multiscale features into our current framework.
- We also plan to add boundary-aware techniques by adjusting the training process, giving the edge pixels more weight and adding edge-based feature maps to enhance the accuracy of segmentation.
- We will try to fit data augmentation strategies such as elastic deformation, noise injection, and contrast variations on the dataset to enhance model generalization. Additionally, we will perform hyperparameter tuning of the Random Forest model, exploring parameters such as the number of trees, maximum depth, and feature subsets to better address class imbalance.
- We will also experiment with more advanced post processing methods such as adaptive morphological operations, better connected component filtering and contour smoothing to produce less noisy and more regular segmentation masks.
- Lastly, we will also apply other performance indices like Hausdorff Distance and precision-recall analysis to get a further understanding of the accuracy of the boundaries and the overall performance of the model.

## References

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